

Multiple Choice

1. **Answer:** A

Options: A) False, True; B) False, False and x is less than 100; C) False, True and x is an odd number; D) True, True and x is greater than 100; E) False, True and x is a prime number

Note: Statement 1 is false because the set $\{1, 2, \dots, n-1\}$ forms a group under multiplication modulo n only if n is prime, while Statement 2 is true as the integer x exists (specifically $x = 63$) such that $63 \times 1 \pmod{100}$, indicating that 63 has a multiplicative inverse modulo 100.

2. **Answer:** A

Options: A) 0.0; B) Undefined; C) 2; D) $n-1$; E) $n+1$

Note: Since the rank of the product of two matrices cannot exceed the rank of either matrix, and given that $\text{rank}(A) = n$ and $\text{rank}(A*B) = 0$, it follows that $\text{rank}(B)$ must be 0, indicating that B maps all vectors to the zero vector.

3. **Answer:** A

Options: A) $e/2$; B) 1; C) e^2 ; D) $2/e$; E) $1/2$

Note: The correct answer is $e/2$ because the function $y = (\ln x)/x$ reaches its maximum value at $x = e$, and substituting this value into the function yields a maximum of $(\ln e)/e = 1/e$, which leads to the overall maximum value of $e/2$ when considering the transformation of the function.

4. **Answer:** A

Options: A) 2.5; B) -2.0; C) 3.0; D) 2.0; E) 1.5

Note: The correct answer is 0, as it is impossible for a holomorphic function defined on the open unit disk to achieve every complex value in $\mathbb{C}$ due to the open mapping theorem, which states that a non-constant holomorphic function maps open sets to open sets, and thus cannot cover the entire complex plane.

5. **Answer:** C

Options: A) 42; B) 14; C) 35; D) 21; E) 49

Note: The correct answer is 35 because applying Fermat's Little Theorem allows us to simplify the calculation of $6^{83}49$ and $8^{83}49$ to $6^{83} \equiv 6^{35}49$ and $8^{83} \equiv 8^{35}49$, leading to the final result 35 when summed and taken modulo 49.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The sequence 17,400, 116,633, 161,455, 167,213 is correctly ordered from least to

greatest because each subsequent number is greater than the previous one, maintaining an increasing order.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the greatest common divisor (GCD) of 120 and 40 is 40, which has 8 positive factors (1, 2, 4, 5, 8, 10, 20, and 40).

8. **Answer:** True

Options: A) True; B) False

Note: The improper fraction representation of a mixed number is found by multiplying the whole number by the denominator and adding the numerator, so for 4 and $\frac{4}{5}$, this results in $(4 * 5) + 4 = 20 + 4 = 24$, which when expressed as an improper fraction is $\frac{24}{5}$, thus the statement is false.

9. **Answer:** False

Options: A) True; B) False

Note: The total cost of a 6-lb bag of cat food at $.47/lb$ is 2.82, which is calculated by multiplying the weight (6 lbs) by the price per pound (\$.47), therefore the statement is false.

10. **Answer:** False

Options: A) True; B) False

Note: The number 1,449 rounds down to 1,400 and does not round to 1,500, while 1,549 rounds up to 1,600, making the statement false.

Long Form Response

11. **Answer:** To determine the domain of the function $f(x) = \frac{x+6}{\sqrt{x^2-3x-4}}$, we must analyze the constraints imposed by the denominator, specifically the square root. The expression under the square root, $x^2 - 3x - 4$, must be positive to ensure that $f(x)$ is defined. We factor this quadratic to find its roots: $(x - 4)(x + 1) = 0$, leading to roots at $x = -1$ and $x = 4$. The intervals where the quadratic is positive are found by testing the sign of the expression in the intervals determined by these roots: $(-\infty, -4)$ and $(4, \infty)$. Thus, the domain of $f(x)$ is $(-\infty, -4) \cup (4, \infty)$.

Note: correct because it accurately identifies that the square root in the denominator must be positive, leading to the analysis of the quadratic expression $x^2 - 3x - 4$ and determining the intervals where the function is defined by finding the roots and testing the sign of the resulting factors.

12. **Answer:** To determine how far a 30 N force will stretch the spring beyond its natural length, we start by using Hooke's Law, which states that the force exerted by a spring is proportional to its extension. Given that it takes 2 J of work to stretch the spring from 30 cm to 42 cm (a stretch of 12 cm), we can calculate the spring constant k using the work done formula $W = \frac{1}{2}kx^2$, where x is the extension. Here, $W = 2 \text{ J}$ and $x = 0.12 \text{ m}$, leading to $k = \frac{2}{0.5 \times 0.12^2} = 277.78 \text{ N/m}$. Now, applying Hooke's Law

$F = kx$ for a force of 30 N, we find $x = \frac{F}{k} = \frac{30}{277.78} \approx 0.108 \text{ m}$ or 10.8 cm. Therefore, the total stretch from the natural length is approximately 13.5 cm, considering the initial extension.

Note: The answer correctly applies Hooke's Law and the work done on a spring to determine the spring constant, which allows for the calculation of how far a specific force will stretch the spring beyond its natural length.

End of Answer Key